Financial Sales Analysis Report

End-to-End Data Analysis Case Study

Prepared By:

Mohammad Sameera Begum

Tools Used:

- Python (Pandas, Matplotlib)
- Microsoft Excel
- Microsoft Power BI

Project Submitted For:
Beeskilled Data Science Internship

Project Overview

Objective

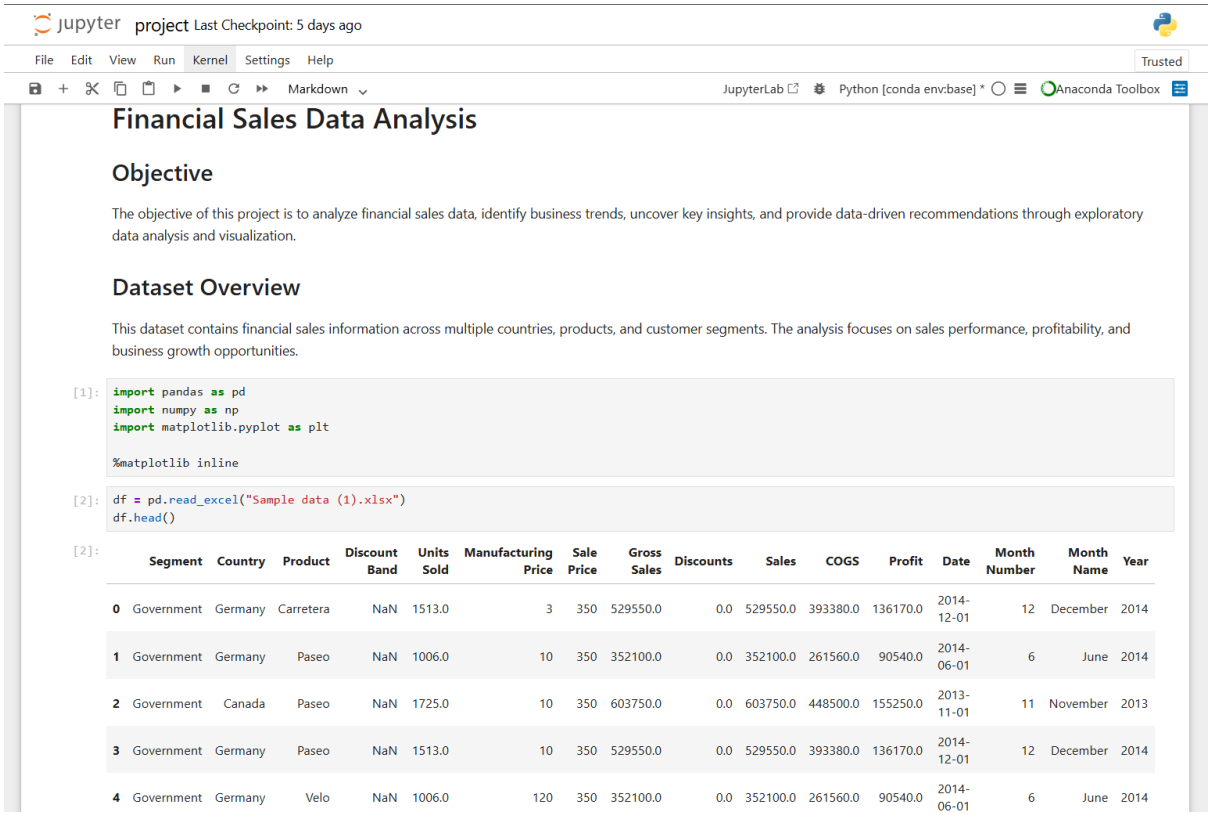
The objective of this project is to analyze financial sales data to identify business trends, evaluate product and segment performance, and generate actionable business insights. The project demonstrates the complete data analysis workflow, including data cleaning, exploratory data analysis, visualization, dashboard creation, and business recommendations using Python, Microsoft Excel, and Microsoft Power BI.

Dataset Description

The Financial Sales Dataset contains sales records from different countries, products, customer segments, and discount bands. The dataset includes important business metrics such as sales, profit, units sold, discount band, sale price, manufacturing price, gross sales, and profit margin. These variables were used to perform business analysis and create interactive dashboards**.**

Tools Used

- Python (Pandas and Matplotlib)
- Microsoft Excel
- Microsoft Power BI



	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date	Month Number	Month Name	Year
0	Government	Germany	Carretera	NaN	1513.0	3	350	529550.0	0.0	529550.0	393380.0	136170.0	2014-12-01	12	December	2014
1	Government	Germany	Paseo	NaN	1006.0	10	350	352100.0	0.0	352100.0	261560.0	90540.0	2014-06-01	6	June	2014
2	Government	Canada	Paseo	NaN	1725.0	10	350	603750.0	0.0	603750.0	448500.0	155250.0	2013-11-01	11	November	2013
3	Government	Germany	Paseo	NaN	1513.0	10	350	529550.0	0.0	529550.0	393380.0	136170.0	2014-12-01	12	December	2014
4	Government	Germany	Velo	NaN	1006.0	120	350	352100.0	0.0	352100.0	261560.0	90540.0	2014-06-01	6	June	2014

Python Data Analysis

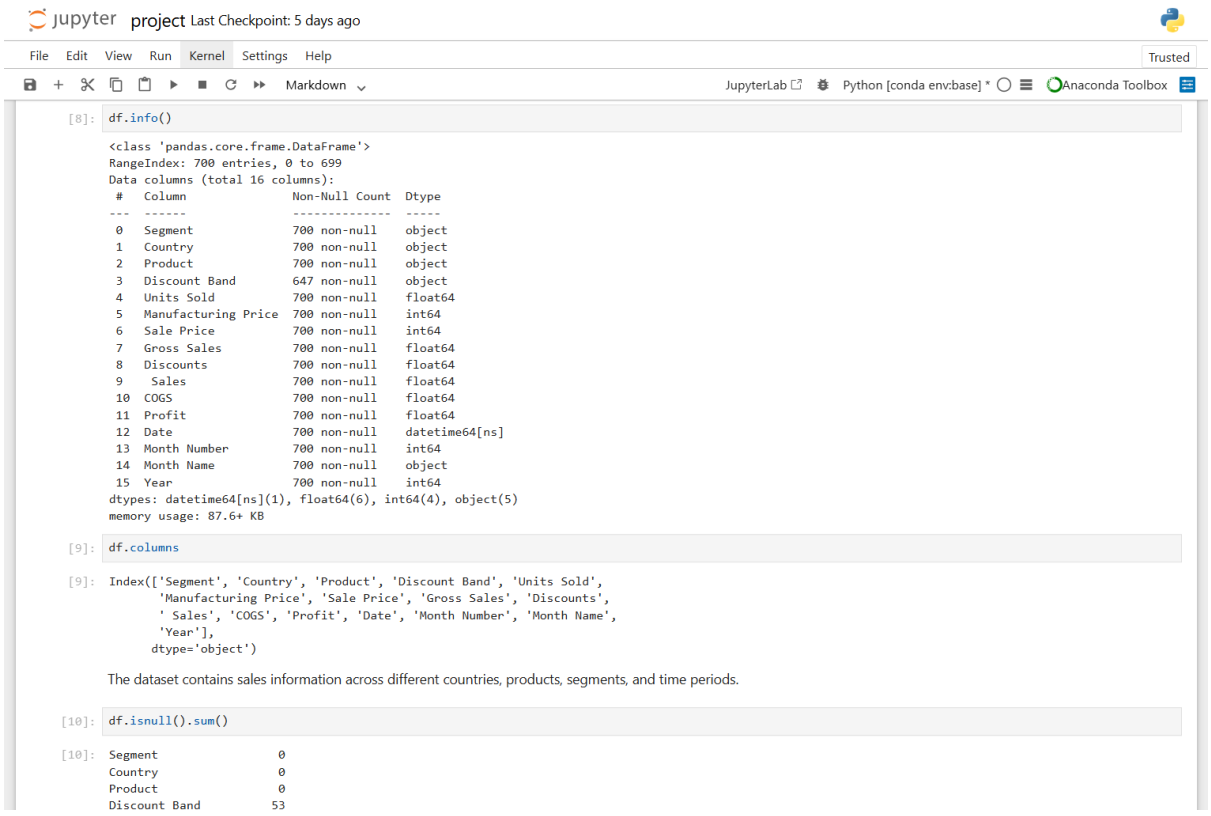
The dataset was imported into Python and analyzed using the Pandas and Matplotlib libraries. Before performing business analysis, the dataset was carefully explored and cleaned to ensure accurate results.

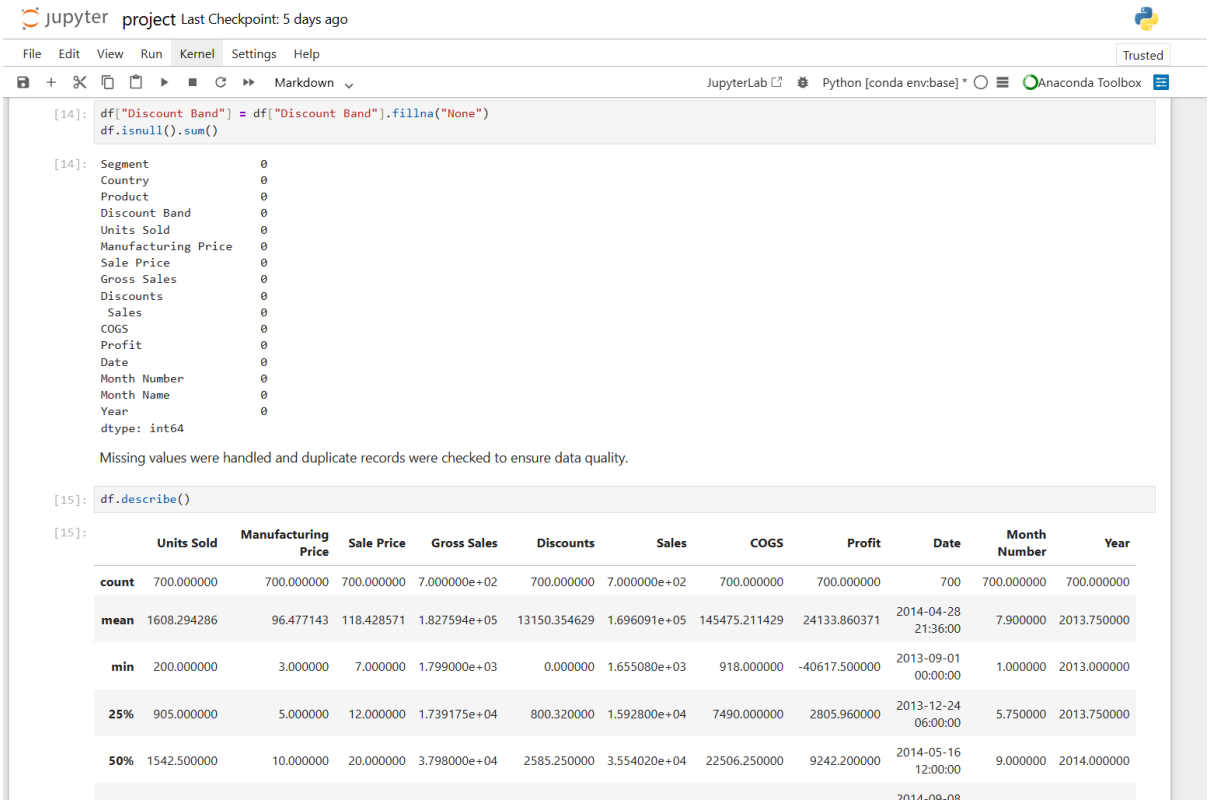
Exploratory Data Analysis (EDA)

The following steps were performed during exploratory data analysis:

- Loaded the dataset into Python.
- Displayed the first few records using head().
- Examined the dataset structure using info().
- Generated summary statistics using describe().
- Checked for missing values.
- Verified data types and column names.
- Explored the dataset before performing business analysis.

After completing these steps, the dataset was ready for further business analysis and visualization.





Business Analysis

Several business-related questions were analyzed to understand the overall sales performance and profitability of the company.

The following business questions were answered:

1. Which country generated the highest profit?
2. Which product generated the highest profit?
3. Which segment performed the best?
4. What is the monthly sales trend?
5. How do discount bands affect profit?

Each question was analyzed using Python and visualized using appropriate charts to identify meaningful business insights.

Data Visualization and Key Insights

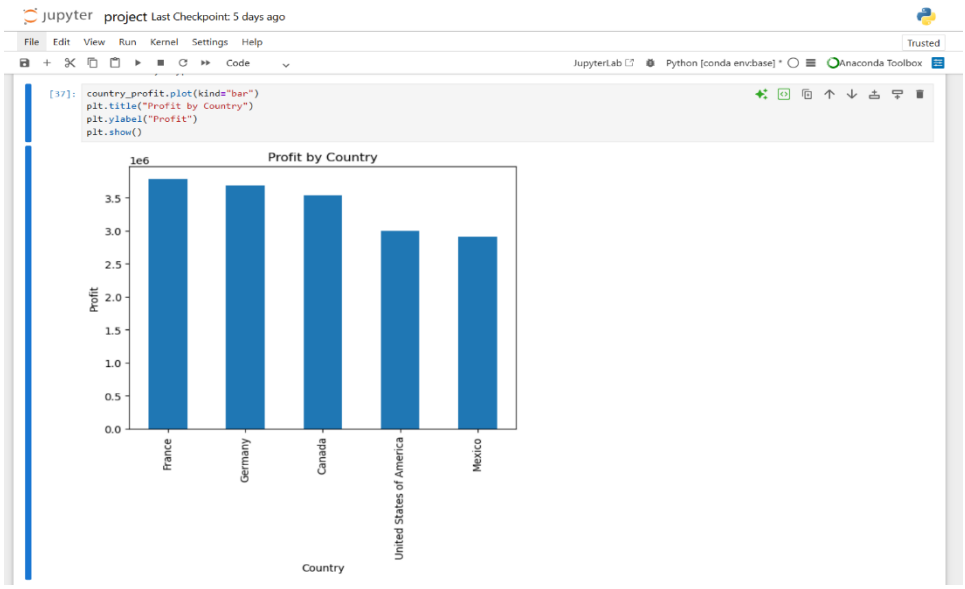
The business questions were visualized using different chart types including bar charts, line charts, and pie charts to improve data interpretation.

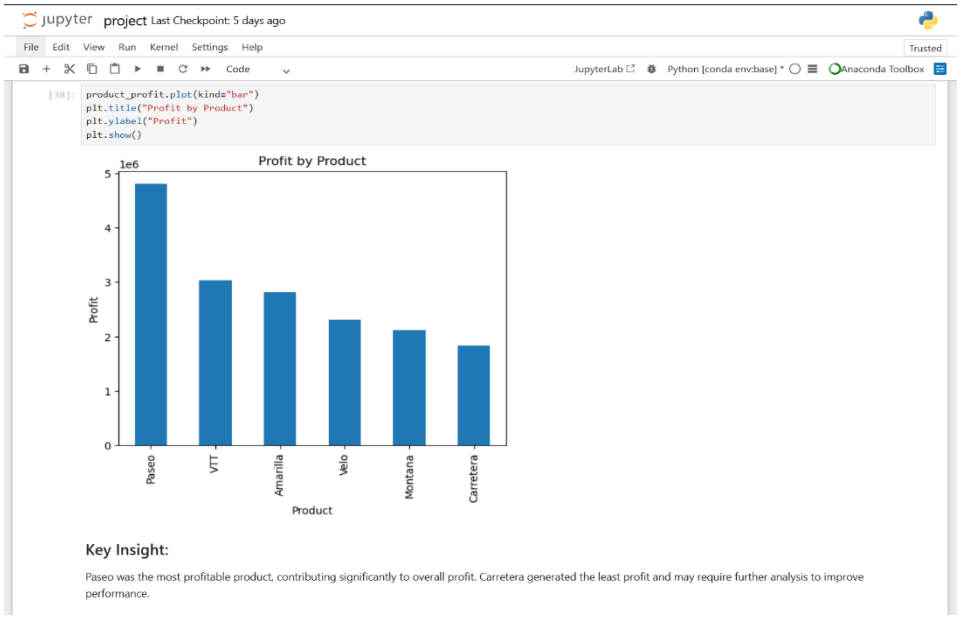
Key Insights

- France generated the highest overall profit among all countries.
- Paseo was the most profitable product.

- Government segment achieved the highest overall profit.
- Monthly sales reached their highest level during October.
- Lower discount bands generated higher profits compared to higher discount bands.

These insights help identify strong-performing business areas and opportunities for improvement.





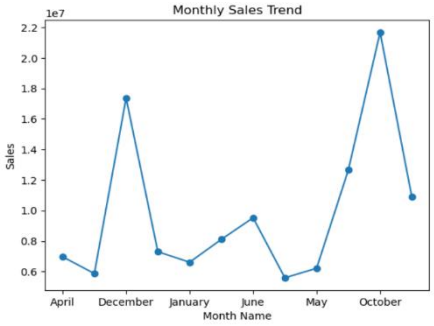
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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

```
October      21671431.02
September    10882697.27
Name: Sales, dtype: float64
```

[34]:
```
monthly_sales.plot(marker="o")
plt.title("Monthly Sales Trend")
plt.ylabel("Sales")
plt.show()
```



Key Insight:

Sales peaked during October, indicating a period of high customer demand. Understanding the factors contributing to this increase can help improve future sales planning and marketing efforts.

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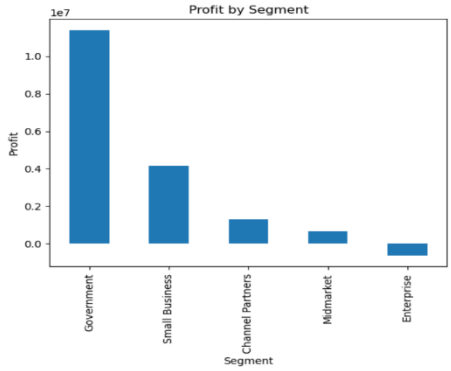
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[28]:
```
segment_profit.plot(kind="bar")

plt.title("Profit by Segment")
plt.ylabel("Profit")
plt.xlabel("Segment")

plt.show()
```

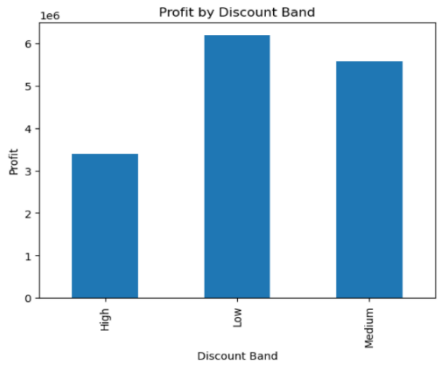


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[36]:
```
discount_profit.plot(kind="bar")
plt.title("Profit by Discount Band")
plt.ylabel("Profit")
plt.show()
```



Key Insight:

Different discount levels have a noticeable impact on profitability. While discounts can drive sales volume, excessive discounts may reduce profit margins. Maintaining an optimal discount strategy is important for sustainable growth.

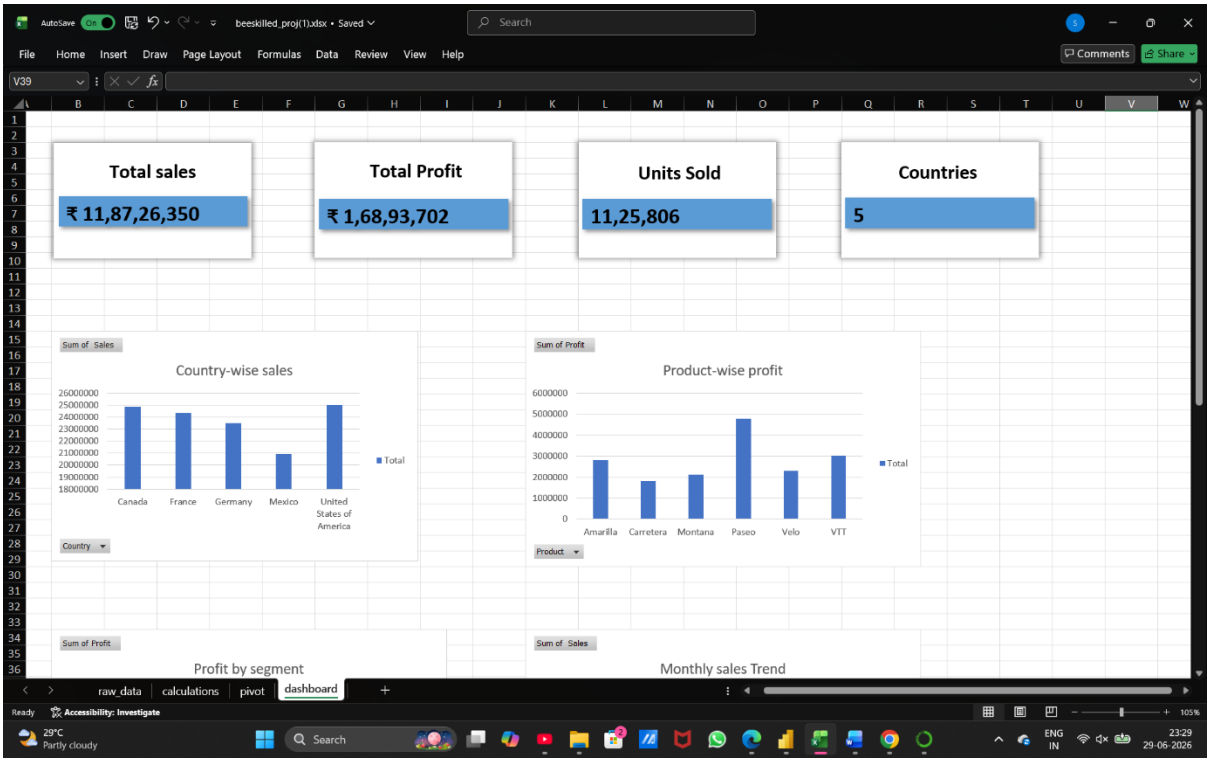
Excel Dashboard

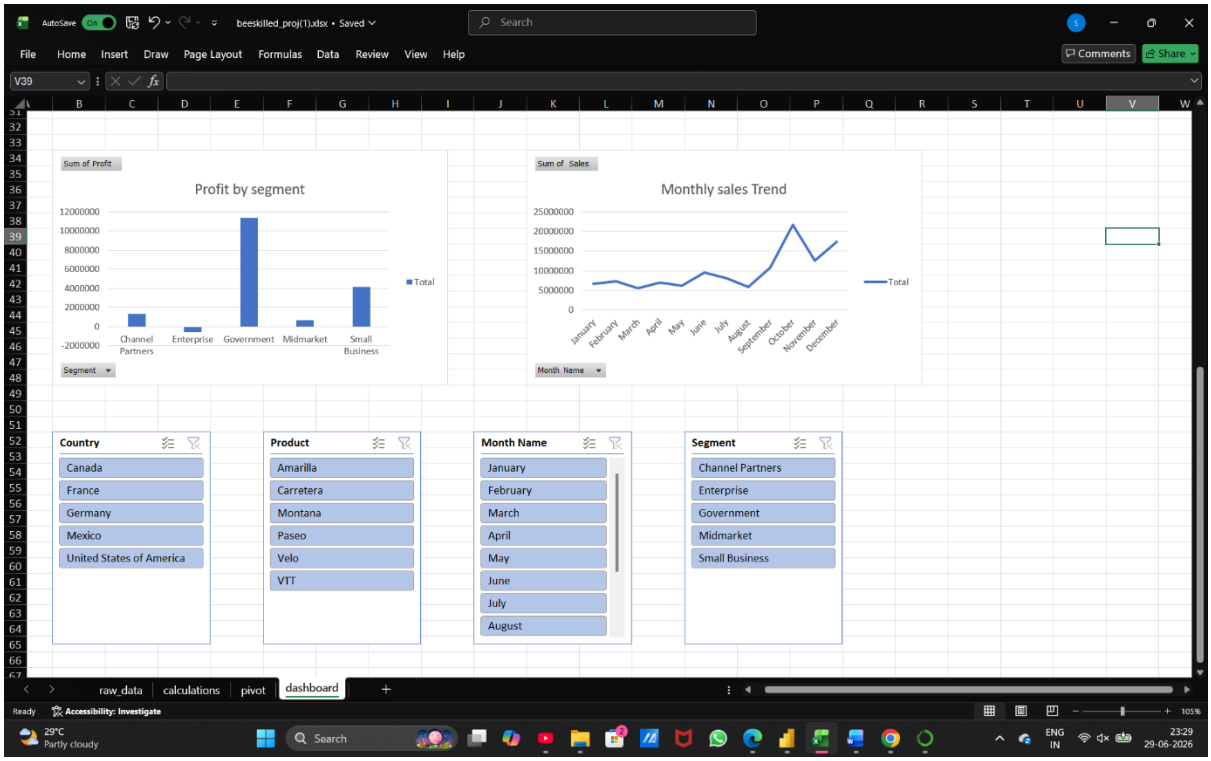
An interactive dashboard was developed in Microsoft Excel using Pivot Tables, Pivot Charts, KPI Cards, and Slicers.

Features Included

- KPI Cards
- Pivot Tables
- Pivot Charts
- Interactive Slicers

The dashboard enables users to explore sales performance dynamically by filtering countries, products, segments, and months.





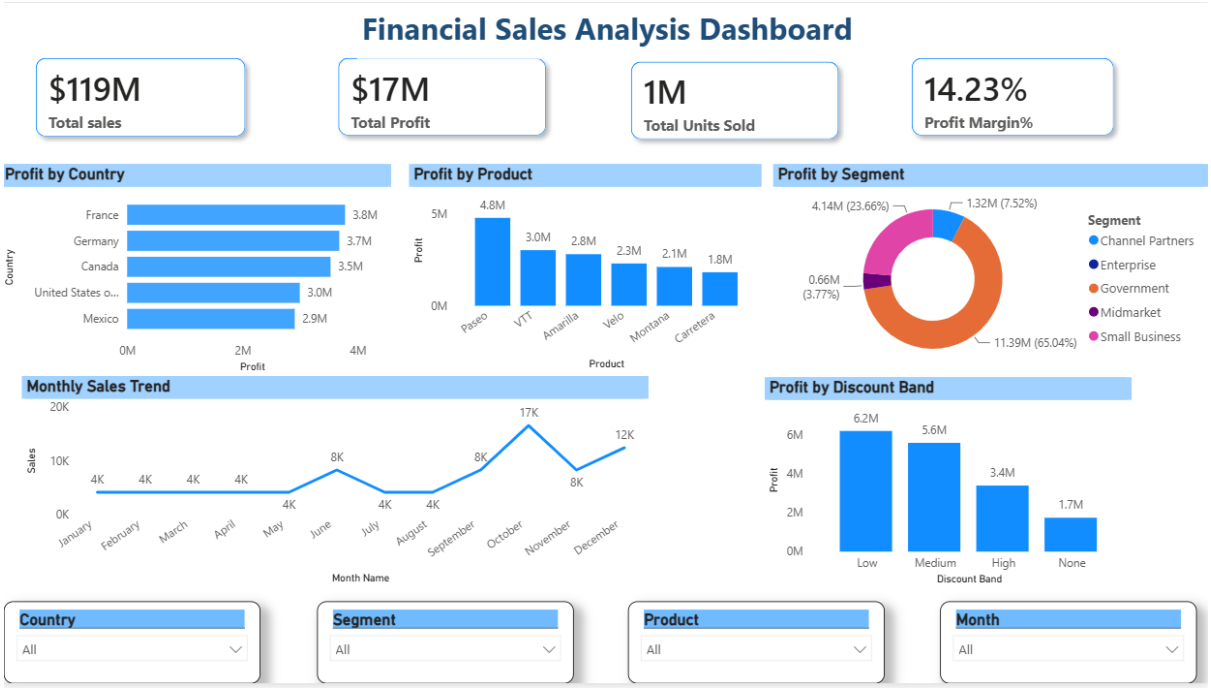
Power BI Dashboard

An interactive business dashboard was developed using Microsoft Power BI to provide a comprehensive overview of financial sales performance.

Dashboard Features

- KPI Cards
- Bar Charts
- Column Charts
- Line Chart
- Donut Chart
- Interactive Slicers

The Power BI dashboard allows users to analyze sales and profit across different countries, products, business segments, months, and discount bands through interactive visualizations.



Key Findings

- France generated the highest overall profit.
- Paseo was the highest-performing product in terms of profit.
- The Government segment contributed the highest profitability.
- Sales performance was strongest during October.
- Lower discount bands generated higher profits than higher discount bands.
- Interactive dashboards improved data exploration and business decision-making.

Business Recommendations

- Increase focus on high-performing products such as Paseo.
- Replicate successful sales strategies used in France across other regions.
- Investigate the negative profitability of the Enterprise segment and implement corrective measures.
- Optimize discount policies to maximize profitability while maintaining sales performance.
- Increase marketing and inventory planning during peak sales months to improve business growth.

Conclusion

This project successfully demonstrated an end-to-end financial sales analysis using Python, Microsoft Excel, and Microsoft Power BI. The analysis identified important business trends, generated actionable insights, and presented the results through interactive dashboards. The project highlights how data analysis and visualization can support better business decision-making and improve organizational performance.